1.1 A Prime Factory

WORKSHEET

The question

STEP II 2002, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The last line asks about *infinitely many primes*. The middle asks about *common factors*. What could possibly connect "no two Fermat numbers share a factor" with "there are infinitely many primes"?

STAGE 1 Getting a feel for the numbers

Covers: the first four Fermat numbers

Ask yourself Be careful with the tower: 2^{2^n} means $2^{(2^n)}$. Work out 2^n first, for $n = 0, 1, 2, 3$, and only then raise 2 to that power.

STAGE 2 Checking the identity

Covers: $(*)$ for $k = 1, 2, 3$

Ask yourself For $k = 1, 2, 3$ the left-hand side of $(*)$ is a product of one, two and three Fermat numbers. Compute each product and compare it with $F_k - 2$.

STAGE 3 Proving the identity

Covers: $(*)$ for every $k \geq 1$

Ask yourself Induction is the natural tool. If $(*)$ holds for k, what do you multiply both sides by to get the left-hand side for $k + 1$?

Ask yourself After that step you need $(F_k - 2)\, F_k = F_{k+1} - 2$. Write $F_k - 2$ and F_k as powers of 2 minus or plus 1.

Pause and try $(a-1)(a+1) = a^2 - 1$. Take $a = 2^{2^k}$: what is a^2 as a power of 2?

STAGE 4 No common factors

Ask yourself Suppose d divides both F_m and F_n, with $m < n$. Use $(*)$ with $k = n$. Is F_m one of the factors on its left-hand side?

Pause and try So d divides $F_n - 2$ as well as F_n. What must d then divide? Which of the possibilities can you rule out, and why?

STAGE 5 Infinitely many primes

Ask yourself Each Fermat number has at least one prime factor. Choose one prime factor of each.

Pause and try Could the prime you chose for F_m be the same as the one you chose for F_n? Use the previous part.

What's really going on

Was Fermat right?

No. In 1732 Leonhard Euler showed that

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

Euler didn't find 641 by luck. He proved that any prime factor of F_n must have a very special form (for $n = 5$, it must be one more than a multiple of 64), which drastically cut down the candidates. You can prove this yourself in Exercise 4.

Since then, every Fermat number anyone has managed to test, from F_5 up to F_{32}, has turned out to be composite. As far as anyone knows, F_0 to F_4 may be the *only* Fermat primes, but nobody has been able to prove it. It is a genuinely open problem.

The delightful twist is that our proof doesn't care. Prime or composite, the Fermat numbers still produce infinitely many primes.

Fermat primes and constructing polygons

Fermat primes turn up somewhere quite unexpected. The ancient Greeks knew how to construct an equilateral triangle and a regular pentagon with ruler and compass, and notice that $3 = F_0$ and $5 = F_1$. For two thousand years nobody could construct any other regular polygon with a prime number of sides. Then in 1796, aged 19, Carl Friedrich Gauss showed how to construct a regular *17-gon*, and $17 = F_2$.

Gauss went further: a regular N-gon can be constructed with ruler and compass exactly when N is a power of 2 multiplied by *distinct Fermat primes*. (Pierre Wantzel completed the proof in 1837.) So which polygons can be constructed depends on how many Fermat primes there are. You will meet the key calculation behind Gauss's 17-gon in Section 6.2 (STEP II 1991, Question 4), and the full story belongs to *Galois theory*, one of the highlights of an undergraduate degree.

One theorem, many proofs

Why would mathematicians bother proving the same theorem more than once? Because different proofs reveal different things. This proof, usually credited to Christian Goldbach in a 1730 letter to Euler, shows that *any* infinite sequence of pairwise coprime whole numbers greater than 1 forces infinitely many primes. Fermat numbers are just one convenient example. (Exercise 3 gives another.)

It also gives more than Euclid's proof does. The primes $p_0, p_1, \ldots, p_{n-1}$ are n *different* primes, each at most F_{n-1}. So there are at least n primes no bigger than $2^{2^{n-1}} + 1$. That is a *quantitative* statement about how the primes are spread out. It is very weak (the truth is much better), but it is a first step towards one of the great results of nineteenth-century mathematics, the *Prime Number Theorem*, which describes exactly how the primes thin out. Section 1.2 takes the next step.

Try this next

1 **Euclid's version.** Write out Euclid's proof that there are infinitely many primes. In what sense is it "by contradiction", and in what sense is the Fermat-number proof "constructive"?

2 **The last digit.** Show that for $n \geq 2$ the Fermat number F_n ends in the digit 7.

3 **Sylvester's sequence.** Define $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$, so the sequence begins $2, 3, 7, 43, 1807, \ldots$

(a) Show that $a_{n+1} - 1 = a_1 a_2 \cdots a_n$.

(b) Deduce that the terms are pairwise coprime, and hence give another proof that there are infinitely many primes.

(c) HARDER Show that $\dfrac{1}{a_1} + \dfrac{1}{a_2} + \cdots + \dfrac{1}{a_n} = 1 - \dfrac{1}{a_{n+1} - 1}$.

4 **Euler's trick.** *This needs Fermat's Little Theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1$*

$(\mathrm{mod}\ p)$. Suppose a prime p divides F_n.

(a) Show that $2^{2^n} \equiv -1 \pmod{p}$, and hence that $2^{2^{n+1}} \equiv 1 \pmod{p}$.

(b) Let d be the smallest positive integer with $2^d \equiv 1 \pmod{p}$. Show that d divides 2^{n+1} but not 2^n, and conclude that $d = 2^{n+1}$.

(c) Using Fermat's Little Theorem, show that d divides $p - 1$. Deduce that $p \equiv 1 \pmod{2^{n+1}}$.

(d) For $n = 5$, this says any prime factor of F_5 has the form $64k + 1$. Check that 641 does. How many primes of this form are smaller than 641?

5 **A bound on the primes.** Use the argument in *One theorem, many proofs* to show that the nth prime is at most $2^{2^{n-1}} + 1$. How good is this bound when $n = 5$?

Further reading

- **Martin Aigner & Günter M. Ziegler, *Proofs from THE BOOK*** (Springer), Chapter 1, "Six proofs of the infinity of primes". It includes this proof alongside Euclid's and four others, one of which uses topology. One of the best books you can read before university.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press), Chapters 1–2. A beautifully written introduction to primes and factorisation, ideal for the summer before university.
- **Michal Křížek, Florian Luca & Lawrence Somer, *17 Lectures on Fermat Numbers*** (Springer, 2001). Everything about Fermat numbers, from Euler's factorisation to Gauss's polygons. Dip into the early chapters.
- **Ian Stewart, *Why Beauty Is Truth*** (Basic Books), the chapter on Gauss. The story of the 17-gon told for a general reader.

The full guided solution is in Section 1.1:
https://beyondstep.keithmcnulty.org/ch01/fermat-numbers.html